Dhanekula Jaya Prakash

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EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2026	7.79
SRI SHIRDI SAI VIDTANIKETAN E M SCHOOL KADIAM AP(57172)	Class XII (CENTRAL BOARD OF SECONDARY EDUCATION)	2023	94.60%
SRI SHIRIDI SAI VIDHYA NIKETHAN H S RJD, EAST GODAVARI DISTRICT	Class X (BOARD OF SECONDARY EDUCATION,ANDHRA PRADESH)	2021	93.83%

SCHOLASTIC ACHIEVEMENTS

- Secured **State Rank 597** in TS EAMCET 2023 among **2 lakh** students
- Secured **State Rank 252** in EAPCET 2023 among **2.5 lakh** students
- Qualified for **ISI admission** test conducted by Indian Statistical Institute, 2023

PROJECTS

Real-Time AI Quiz Platform | C++, Node.js, Express, React, WebSockets, Google Gemini AI, JWT, MongoDB

- Built a **multi-threaded C++ TCP server** handling concurrent client connections using a custom **9-byte binary application-layer protocol** with byte-stream framing to ensure reliable packet delivery over TCP.
- Implemented a **thread-safe reader-writer cache** using **shared mutex** with TTL-based expiration, reducing redundant Gemini API calls under concurrent load.
- Integrated **Google Gemini 1.5 Flash** for dynamic **AI-generated MCQ trivia** and built a **Node.js Express API gateway** to bridge the C++ engine to a React frontend via binary socket serialization.
- Developed a **real-time WebSocket leaderboard** broadcaster with **JWT authentication** for the clients.

Multi-Media Database System (MMDB) | Python, FastAPI, Streamlit, Celery, Redis, LanceDB, PyTorch, CLIP, CLAP, Qwen3-VL, Faster-Whisper, PaddleOCR, spaCy, PyMuPDF

- Built an **asynchronous multi-modal ingestion pipeline** using **FastAPI**, **Celery**, and **Redis** to process and index heterogeneous assets like Images, Audio, Video, PDFs, TXT without blocking API response times.
- Integrated **SOTA deep learning models** -- **CLIP**, **CLAP**, **Qwen3-VL**, and **Faster-Whisper** -- mapping all media into a **dense vector space** for cross-modal semantic search.
- Built a **hybrid retrieval engine** combining **BM25 keyword** ranking using spaCy and RapidFuzz with **dense vector search** via **LanceDB**, delivering high-precision semantic and exact-match query results.
- Implemented visual processing pipelines using **PyMuPDF** and **PaddleOCR** to extract text layouts, render page images, and process inline graphics from documents into searchable segments.
- Developed an interactive **Streamlit dashboard** for real-time media ingestion and hybrid semantic search, with **lazy-loaded model singletons** to minimize cold-start memory overhead.

Ride Booking Platform | React.js, Node.js, Express, MongoDB, Socket.IO, Google Maps API, JWT, Tailwind CSS, bcrypt

- Developed a **full-stack ride-booking platform** with **JWT authentication**, token blacklisting, and **bcrypt password hashing** for secure session management.
- Implemented **real-time bidirectional communication** using **Socket.IO** to instantly broadcast ride requests to nearby captains and push live status updates back to users.
- Integrated **Google Maps APIs** for Geocoding, Distance Matrix, Places Autocomplete and implemented a **geospatial driver-matching system** using MongoDB's **geoWithin** to find captains within pickup radius.

- Engineered a **dynamic fare calculation engine** for different vehicle types using distance and time-based pricing via **Google Distance Matrix API** with **OTP**-based ride verification.

SKILLS AND EXPERTISE

- **Skills:** DSA, Competitive Programming, Object Oriented Programming, Socket Programming, Web Development
- **Languages:** C, C++, Python, JavaScript
- **Web Technologies:** HTML, CSS, React.js, Node.js, Express.js
- **AI / ML:** PyTorch, Scikit-learn, Pandas, NumPy, Matplotlib, Hugging Face Transformers
- **Databases:** MongoDB, SQL, LanceDB
- **Tools:** Git, GitHub, Docker, Redis, VS Code, Jupyter Notebook, Bash

RELEVANT COURSES

- **Computer Science courses**: Introduction to Programming, Software Development fundamentals, Data Structures, Algorithms, Operating Systems, Introduction to AI, DBMS, Compilers, Computer Architecture, Computer Networks, FOML, Deep Learning, Machine learning for Physical Sciences, Fraud Analytics, Software Engineering.
- **Mathematics and EE courses**: Calculus, Elementary Linear Algebra, Probability and Random Variables, Transform Techniques, Introduction to Statistics, Digital Circuits.

EXTRACURRICULAR

- Captain of the institute Kabaddi team and led the team in many institute tournaments organized by the EBSB Club.
- Actively participated in Volleyball and Hockey events during Milan'23, Milan'24, Diesta'23, and Diesta'24.
- Student Volunteer, National Service Scheme (NSS).